

Composio App Intelligence Research

Empirical Evaluation of 100 SaaS Integrations for AI-Agent Readiness

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Empirical Scope:	100 Active SaaS Profiles	Audit Accuracy:	88.1% (21-App Stratified Audit)
Live Case Study:	https://ishaanpgujaran.github.io/composio-product-research-agent/		
Source Code Repo:	https://github.com/ishaanpgujaran/composio-product-research-agent		

1. Executive Summary & Core Empirical Findings

We deployed an autonomous parallel research operator using Composio tools and Google Gemini to audit 100 SaaS applications for AI-agent readiness. Key empirical distributions discovered across the dataset:

- **Auth Standards:** 60.0% of apps use OAuth 2.0 or Bearer tokens, establishing standard credential lifecycles.
- **Access Gating:** 84.0% provide open self-serve API access, while 11.0% require enterprise sales contact.
- **MCP Prevalence:** 7.0% feature Model Context Protocol servers (official native support remains nascent).
- **Buildability:** 64.0% received an 'Easy' verdict with public OpenAPI specs; 12.0% face operational blockers.

2. Research Operator Pipeline & Model Cascade

To prevent pipeline stalls under Gemini Free Tier daily limits (20 req/day), we engineered a dynamic fallback cascade: Gemini 3.7 Flash -> 3.6 Flash -> 3.5 Flash -> 3.5 Flash Lite -> 3.1 Flash Lite. When an active tier is throttled (HTTP 429), a thread-safe lock automatically routes subsequent queries to the next available tier. This allowed all 100 applications to run continuously while Pydantic v2 schemas enforced strict data contracts.

3. Ground Truth Verification & Human-in-the-Loop Audit

A stratified 21-app sample was audited across 8 core fields, confirming 88.1% accuracy (148/168 correct fields). In our commitment to full transparency, all pipeline errors and human corrections are highlighted directly:

- **Salesforce MCP Correction:** Agent extracted 'Community' via Smithery -> Corrected to 'Official' hosted servers.
- **Slack MCP Correction:** Agent extracted 'Third-party' -> Corrected to 'Official' native support in Slack help docs.

4. Setup & Local Reproduction Commands

Execute the research pipeline, run ground-truth audits, or serve the dashboard locally:

```
# 1. Setup virtual environment and install dependencies
python3 -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt
# 2. Run batch research operator (multi-model cascade handles quotas automatically)
python -m src.batch --config data/apps.json --output data/pass_1.json
# 3. Run ground truth verification & serve dashboard at http://localhost:8000
python -m src.verify --sample && python -m src.verify --run && python3 -m http.server 8000
```